# Huawei Access Node WFD-ACC-09 Optical Transport Failure SOP (ALARM_6316)

**Node:** WFD-ACC-09 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_6316

## Description

This Standard Operating Procedure (SOP) outlines the steps to diagnose and resolve an Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on the Huawei Access Node WFD-ACC-09 at the Whitefield POP. The failure domain occurs due to the inherent limitations of the Optical Transport Network (OTN) protocol, which relies on the integrity of the optical signal transmitted over the fiber. In the event of a fiber cut or SFP/XFP degradation, the OTN protocol fails to detect the error, resulting in a loss of signal and subsequent ALARM_6316 alarm. This SOP is designed to quickly identify the root cause of the failure and implement corrective actions to restore network connectivity.

## Pre-requisite Checks

- Pre-requisite: Verify the site location and node ID (WFD-ACC-09) to ensure the correct node is being addressed., Description: This step is crucial to prevent unnecessary troubleshooting and to ensure that the correct node is being targeted.

- Pre-requisite: Check the redundant link status to ensure that the failure is not due to a link failure., Description: A link failure may cause the ALARM_6316 alarm, and verifying the redundant link status helps to rule out this possibility.

- Pre-requisite: Dispatch remote hands to the Whitefield POP to gain physical access to the node., Description: Physical access is required to perform hardware-related tasks, and remote hands may be necessary to facilitate this access.

## Potential Root Causes

| Cause | Probability |
|---|---|
| SFP/XFP degradation | High |
| Fiber cut | Medium |
| Rx/Tx power loss | Low |

## Diagnostic Commands & Expected Outputs

```
Command: display optical-transceiver [interface-name], Expected Output: Optical Transceiver Information, Description: This command displays the optical transceiver information, including the SFP/XFP status and optical signal quality. It helps to identify any issues with the SFP/XFP and the optical signal quality.

Command: display optical-link [interface-name], Expected Output: Optical Link Information, Description: This command displays the optical link information, including the Rx/Tx power level and optical
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```
signal quality. It helps to identify any issues with the optical link and the Rx/Tx power level.

Command: display alarm [ALARM_6316], Expected Output: Alarm Information, Description: This command displays the alarm information, including the alarm code, description, and severity. It helps to verify that the ALARM_6316 alarm is indeed associated with the Optical Transport Failure.
```

## Resolution Steps

Step 1: Step: Replace the SFP/XFP, Command: replace optical-transceiver [interface-name], Description: Replace the SFP/XFP with a new one to resolve the SFP/XFP degradation issue.

Step 2: Step: Verify the fiber connection, Command: display optical-link [interface-name], Description: Verify the fiber connection to ensure that it is secure and not damaged.

Step 3: Step: Adjust the Rx/Tx power level, Command: set optical-link [interface-name] rx-power [value] tx-power [value], Description: Adjust the Rx/Tx power level to resolve the Rx/Tx power loss issue.

## Post-Check Verification

```
Command: display optical-transceiver [interface-name], Expected Output: Optical Transceiver Information, Description: Verify that the SFP/XFP status and optical signal quality are normal after the resolution steps.

Command: display optical-link [interface-name], Expected Output: Optical Link Information, Description: Verify that the Rx/Tx power level and optical signal quality are normal after the resolution steps.

Command: display alarm [ALARM_6316], Expected Output: Alarm Cleared, Description: Verify that the ALARM_6316 alarm is cleared after the resolution steps.
```

## Roll-back Procedures

- Step: Restore the original SFP/XFP, Command: restore optical-transceiver [interface-name], Description: Restore the original SFP/XFP in case the replacement SFP/XFP does not resolve the issue.

- Step: Re-verify the fiber connection, Command: display optical-link [interface-name], Description: Re-verify the fiber connection to ensure that it is secure and not damaged.

- Step: Re-adjust the Rx/Tx power level, Command: set optical-link [interface-name] rx-power [value] tx-power [value], Description: Re-adjust the Rx/Tx power level to the original value in case the adjustment does not resolve the issue.